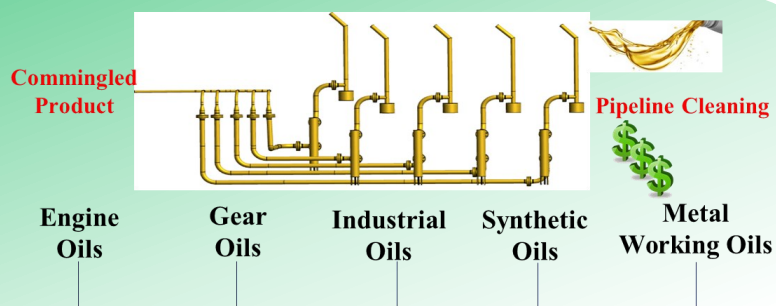


Roadmap for Efficient Transportation and Packaging Processes in Petroleum Pipelines

Through support from the U.S. Environmental Protection Agency Pollution Prevention Program, Rowan University has developed an integrated experimental and mathematical modeling approach to simulate and enhance cleaning efficiency in multi-product petroleum pipelines.



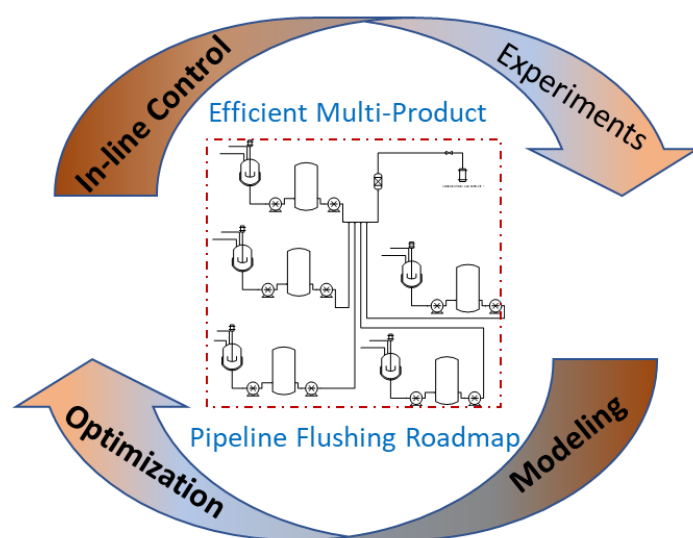
Background

Cleaning of pipelines between product changeovers is crucial for maintaining the quality and purity of oil products in multi-product pipeline systems¹. Existing cleaning methods are inefficient and lead to formation of low value mixed oils. Our goal is to model the cleaning operation and estimate operational regions to minimize the oil downgrade.

Solution Approach

- ♦ **Pilot Plant Development:** Built an industrial scaled down pilot plant to refine the cleaning procedure and model the cleaning operations²
- ♦ **Mathematical Model Development:** Developed mathematical models³ to simulate the flushing process for two operational pipeline scenarios of
 - (i) Direct Oil –to– Oil cleaning process
 - (ii) Air-assisted cleaning process

Framework



Projected Outcomes

- ♦ **Projected Improvements:** Improved models anticipate **~70%** improvement in industrial operations
- ♦ **Benchmark:** Previous model¹ demonstrated an **18%** improvement at industrial scale
- ♦ **Enhanced Cleaning:** Air-assisted cleaning reduces flushing volume by **~30%**



Developed Pilot Plant

The developed **roadmap and physics-driven insights** guides optimal operations, ensuring **product specifications, minimizing oil use**, and improving **overall productivity**

The developed physics-based flushing model simulates the lubricant changeovers by capturing key factors such as **pipeline geometry**, **fluid properties**, and **flow conditions**. It predicts the full transition during flushing and determines the **optimal flush time** and **volume** needed to reach the desired product quality. Using inputs such as **viscosity**, **density**, **flow rate**, **system volume**, and **purity tolerance**, the model replaces *trial-and-error* with *physics-informed* approach for **smarter** and more **efficient** lubricant pipeline operations.

Industry-based lubricant oil classification

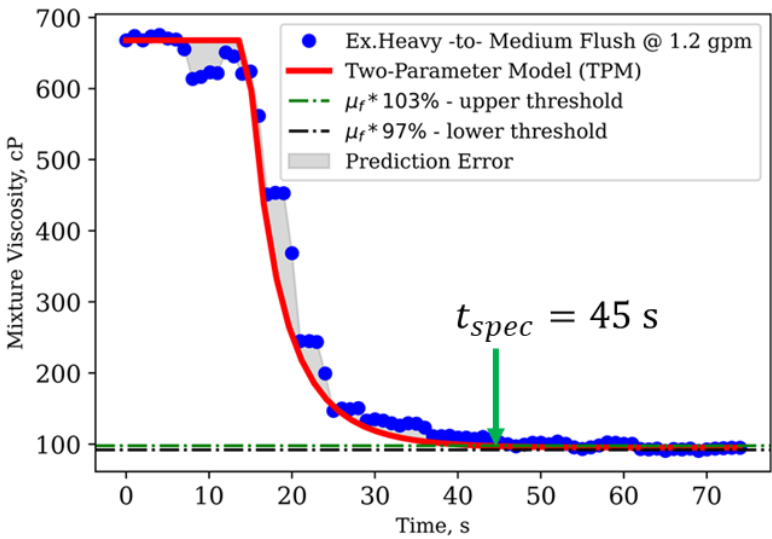
Category	Viscosity (cSt) at 40 °C
Light (L)	0 - 39
Medium (M)	40 – 99
Heavy (H)	100 – 299
Extra Heavy (EH)	300+

Oil categories explored under the study

Previous Product	Next Product			
	L	M	H	EH
	L			
	M			
	H			
EH				

Sample case study:

Changeover from an EH oil (320 cSt) –to– M (47 cSt) oil



Physics-based flush model (red line) validated with experimental observations (blue dots) with prediction of optimal flush time, t_{spec}

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List of Publications

(1) S. S. Jerpoth, R. P. Hesketh, C. S. Slater, M. J. Savelski, and K. M. Yenkie, "Computational Modeling of Lube-Oil Flows in Pipelines to Study the Efficacy of Flushing Operations," in *Computer Aided Chemical Engineering*, vol. 49, Y. Yamashita and M. Kano, Eds., in 14 International Symposium on Process Systems Engineering, vol. 49, Elsevier, 2022, pp. 895–900. doi: 10.1016/B978-0-323-85159-6.50149-4.

(2) B. Gao S.S. Jerpoth, S. Curtis, D. Theuma, S. Roth, M. Fracchiolla, E. Padros, R. P. Hesketh, C. S. Slater, and K. M. Yenkie, "Improved Design of Flushing Process for Multi-Product Pipelines," presented at the Foundations of Computer-Aided Process Design, Breckenridge, Colorado, USA, Jul. 2024, pp. 137–144. doi: 10.69997/sct.171679.

(3) Hesketh, R. P., Gao, B., Yenkie, K. M., Jerpoth, S. S., Theuma, D. A., Curtis, S., Fracchiolla, M., Slater, C. S., Savelski, M. J., & Roth, S. (2024). A Novel Laboratory-Scale Pilot Plant Study. *2024 ASEE Annual Conference & Exposition*. <https://peer.asee.org/46468.pdf>

(4) Jerpoth, S. S., Hesketh, R., Slater, C. S., Savelski, M. J., & Yenkie, K. M. (2023). Strategic Optimization of the Flushing Operations in Lubricant Manufacturing and Packaging Facilities. *ACS Omega*, 8(41), 38288–38300. <https://doi.org/10.1021/acsomega.3c04668>

(5) Jerpoth, S. S., Gao, B., Aboagye, E. A., Hesketh, R. P., Slater, C. S., & Yenkie, K. M. (2024). A Machine Learning Approach for Optimizing Flushing Operations in Lube Oil Manufacturing and Packaging Plants.